

Testing High-Speed Memories

High-speed Flash memory and solidstate drives (SSDs) have become a staple diet of all consumer electronics devices from smartphones to laptop computers. It is almost impossible to imagine a gadget without high-speed storage. Thus it is obvious that a vital part of the production and testing of these devices is to ensure that their memory matches in speed with the rest of the device



(Image courtesy: www.home.agilent.com)

ASHWIN GOPINATH

A recent survey by the International Electronics Manufacturing Initiative (iNEMI) asked test engineers in the electronics industry what their biggest problems were while testing circuit boards.

Of the eleven possible problems listed, characterising and testing memories soldered to circuit boards was among the top three. Memory test topped the list of prevalent problems along with 'lack of access to test points' and 'the need to perform debug/diagnostics on board failures.' Clearly, the ability to thoroughly characterise, test and diagnose problems

with soldered-down memories is one of the most pressing problems in the industry.

When memory speeds were not as high as today's communication protocols' and over memory buses were not as complex, performing static shorts and opens testing on memory interconnects might have sufficed. Today, signal propagation through passive devices such as capacitors and signal integrity on high-speed traces to memory must be validated and characterised for an open data window. Often this data window demonstrates sensitivities to clock jitter, temperature, electrical noise as well as the level and stability of the voltage.

One of the several factors that have exacerbated the difficulties in memory testing is the complexities of these buses. Many of the prominent memory buses, like the various generations of double data rate (DDR) memory, have achieved extremely high data transfer rates at the expense of simplicity. Indeed, the difficulties in testing memory buses have only been exacerbated by each successive generation's high transfer speeds.

Under the hood of memory testing

All memory tests are based on writing to and reading from the memory. Test reads and writes conducted from the board's functional or normal memory access unit in a normal operating manner can be referred to as functional memory tests. Usually, functional memory tests can prove out the design or functionality of a given memory architecture, but these may not be able to detect or even diagnose manufacturing or assembly defects.

Most manufacturing memory tests are algorithmic and structural pseudo-functional tests. The most prevalent form of an algorithmic pseudo-functional memory test is memory built-in self-test (MBIST). MBIST implies that the test is implemented on-silicon or provided by an algorithmic test instrument embedded within a memory test system.

Since memory testing is an altogether different field of operation, parameters tested and methods for testing vary a great deal.

Explaining the memory test, Abhay Samant, marketing manager, National Instruments, says, "A memory test